

Dessin D'enfants

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Why Do We Care?

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However, there is no simple explicit description of a general element of Γ .

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Then, we must choose: $\sigma(\sqrt{1 + \sqrt{2}}) = \pm\sqrt{1 - \sqrt{2}}$.

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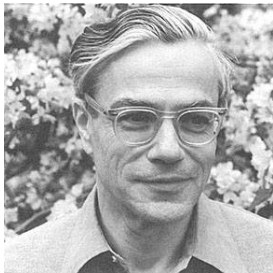
Hence, we would like a different way to study Γ .

(Potential) Resolution

We all learned about u -sub in Calc-II: it does not provide the solution for us straightaway, but it really provides a clearer picture of the problem.

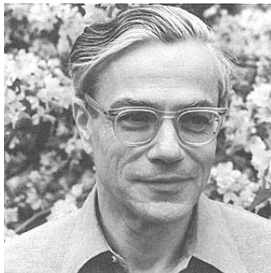
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Analogously, instead of studying Γ itself, Grothendieck decided to interpret the action of Γ on “Dessin D’enfants”.

Definition of Dessin

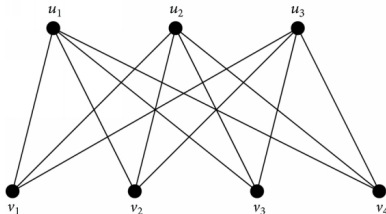
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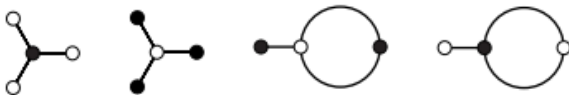
Definition: (Bigraph)

A bigraph is a connected bipartite graph with a fixed two coloring into nonempty sets of black & white vertices. (i.e. no adjacent vertices share the same color.)



Definition of Dessin

A Dessin D'enfant is a pair (X, D) where X is a compact oriented topological surface and D is a finite bipartite graph embedded in X such that $X \setminus D$ is a union of open disks (i.e. the faces of Dessin).



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- Dessin connects algebraic geometry and Galois theory mainly through “Belyi's Theorem”.
- Belyi's theorem is derived from certain maps between algebraic curves.
 - These algebraic curves are defined over the algebraic numbers.
- Grothendieck showed that:

Dessins $\iff$ maps derived from Belyi's theorem

Example: Tree & Sphere

A tree contains no cycles: when we embed trees into a sphere, we will only have one face.

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(The white vertices is a combinatorial tool to ensure bipartiteness)

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Let X be a compact Riemann surface, we will take Chow's theorem for granted, which asserts that:

compact Riemann surface $\longleftrightarrow$ smooth projective algebraic curve over $\mathbb{C}$.

Theorem: (Belyi's Theorem)

- X defined over $\overline{\mathbb{Q}}$ if and only if $\beta : X \rightarrow \hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ is a ramified covering map with branch values contained in $\{0, 1, \infty\}$.
- (i.e. β is the “Belyi map” - the definition is a consequence of the above theorem.)

Definition: (Belyi Pair)

- (X, β) : where X is a compact Riemann surface, β is the Belyi map.

Belyi Map & Dessin

The LHS of BT suggests that our Riemann Surface / Algebraic Curve imposed with this algebraic structure, essentially has a topological correspondence.

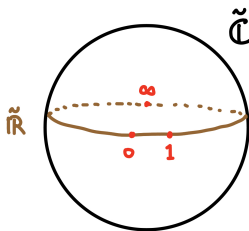
The Belyi pair encodes the topological data associated with the covering.

From a Belyi Map to a Dessin

Let $\beta : X \rightarrow \widehat{\mathbb{C}} \cong \mathbb{P}^1(\mathbb{C})$ be a Belyi map.

The three distinguished points $0, 1, \infty$ divide $\mathbb{P}^1(\mathbb{R}) = \mathbb{R} \cup \{\infty\}$ into three arcs

$$[0, 1], \quad [1, \infty], \quad [\infty, 0]$$



These three arcs form the boundaries of two triangles on the Riemann sphere, with vertices $0, 1, \infty$.

From a Belyi Map to a Dessin

Now, pulling back this triangulation along β gives a triangulation of X .

On $\widehat{\mathbb{C}}$	On X
$0, 1, \infty$	$\beta^{-1}(0), \beta^{-1}(1), \beta^{-1}(\infty)$
three arcs	their preimages under β
two triangles	many lifted triangles

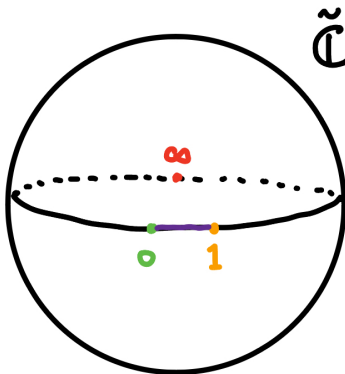
The dessin keeps only

$$D = \beta^{-1}([0, 1]),$$

with $\beta^{-1}(0)$ colored black and $\beta^{-1}(1)$ colored white.

Arise of Dessin from Belyi Map

Hence, we obtain a bipartite graph embedded in X as roughly illustrated below:



Arise of Dessin from Belyi Map

Proposition: (Construction of Dessin)

- Let β be a Belyi map for some X .
- The bipartite graph with black vertices $\beta^{-1}(0)$, white vertices $\beta^{-1}(1)$, and edges $\beta^{-1}([0, 1])$ is a dessin embedded in X .

Example

Let $X = \hat{C}$ and $\beta(x) = x^2$, then let's consider the behaviour of the preimage of 0, 1.

$\beta^{-1}(0) = 0$, with multiplicity 2, then it is a black vertex of valency 2.

$\beta^{-1}(1) = \pm 1$, each with multiplicity 1, then they are white vertices of valency 1.

Hence, the corresponding Dessin is:



Galois, Dessin, Belyi

Theorem: The isomorphism classes of Dessins are equivalent to the isomorphism classes of Belyi maps.

We take this for granted - but it is essentially saying that: to study Dessin, it suffices to study the Belyi map.

Galois, Dessin, Belyi

We may write

$$\overline{\mathbb{Q}} = \bigcup_{K/\mathbb{Q} \text{ finite Galois}} K.$$

For every $g \in \Gamma$ and every finite Galois extension $K/\mathbb{Q}$,

$$g|_K \in \text{Gal}(K/\mathbb{Q}).$$

An element $g \in \Gamma$ is determined by its compatible restrictions to all such finite Galois extensions.

Galois, Dessin, Belyi

Let (X, β) be a Belyi pair defined over $\overline{\mathbb{Q}}$.

For

$$\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}),$$

applying σ to the algebraic coefficients defining both X and β produces another Belyi pair

$$(X, \beta) \longmapsto (X^\sigma, \beta^\sigma).$$

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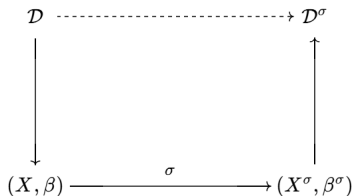
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Since Belyi pairs correspond to dessins, this induces an action of Γ on dessins.

Galois, Dessin, Belyi



This diagram illustrates how the Galois action on a Belyi pair induces an action on its corresponding dessin.

Starting with (X, β) , applying σ gives the conjugate Belyi pair (X^σ, β^σ) , which corresponds to the conjugate dessin $\mathcal{D}^\sigma$.

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Theorem: (Grothendieck) The absolute Galois group

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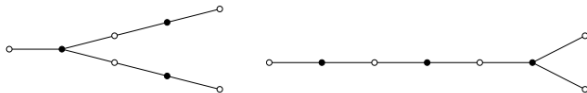
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Thus, dessins provide combinatorial objects through which we can study the action of the absolute Galois group.

A Nontrivial Galois Action on a Dessin



The Belyi map is:

$$f(z) = z^3 \left(z^2 - 2z + \frac{34}{7} \pm \frac{6\sqrt{21}}{7} \right)^2$$

Let

$$\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$$

satisfy $\sigma(\sqrt{21}) = -\sqrt{21}$.

Then σ acts on the coefficients of the Belyi map, interchanging the two conjugate Belyi maps and hence the corresponding dessins.

Thanks!